NATIONAL TRANSPORTATION SAFETY BOARD
Vehicle Recorder Division
Washington, DC 20594

November 3, 2009

Stick Shaker Sound Spectrum Study

Specialist's Study Report
By Christopher Babcock

A. EVENT

Location:	Clarence Center, New York
Date:	February 12, 2009, 2217 Eastern Standard Time (EST)[1]
Aircraft:	Dash 8-Q400, N200WQ
Operator:	Colgan Air Inc. d.b.a. Continental Connection flight 3407
NTSB Number:	DCA09MA027

B. GROUP

No group was formed.

C. ACCIDENT DESCRIPTION

On February 12, 2009, about 2217 EST, a Colgan Air Inc., Bombardier Dash 8-Q400, N200WQ, d.b.a. Continental Connection flight 3407, crashed during an instrument approach to runway 23 at the Buffalo-Niagara International Airport, Buffalo, New York. The four flight crew and 45 passengers were fatally injured and the aircraft was destroyed by impact forces and post crash fire. There was one ground fatality. Night visual meteorological conditions prevailed at the time of the accident. The flight was a Title 14 Code of Federal Regulations (CFR) Part 121 scheduled passenger flight from Liberty International Airport, Newark, New Jersey, to Buffalo.

D. DETAILS OF INVESTIGATION

Analysis of the CVR, FDR, and aircraft performance data indicate in apparent inconsistency in the recorded stall protection system (SPS) data. Specifically, FDR data note that the stick shaker component of the SPS is active at 22:16:34.62 and remains active for each sample recorded every four seconds until the end of the recording.[2] CVR information from the CVR group transcript indicates the absence of stick shaker activation in the time period of 22:16:34.1 EST to 22:16:35.4 EST.[3]

FDR data recorded during this time indicates that the captain's control column position increases from 1.1º to 7.0º in 0.5 seconds. Captain's control column force

[1] All times are expressed in local EST, unless otherwise noted.
[2] See FDR Group Chairman's Factual Report
[3] See CVR Group Chairman's Factual Report

increases from 9.2 lbs to 35.6 pounds in 2 seconds. The low sampling rate for control column force likely understates the maximum force applied over that duration.

A sound spectrum study was performed in effort to identify any frequency content consistent with stick shaker activation that is present during the period 22:16:34.1 EST through 22:16:35.4 EST that may have been inaudible or unidentifiable to the CVR group. An area of known stick shaker activation was studied in order to obtain time and frequency response information regarding stick shaker operation. Figure 1 shows FDR data overlaid with a spectrogram of CVR data for the same time period. Times of stick shaker operation determined by the CVR group are denoted. A spectrogram is a 3D plot displaying time on the x-axis, frequency on the y-axis from 0-8000 Hz, and sound intensity in color. It is a simple way to show differences in intensity with respect to time and frequency.

Two methods were employed to determine if frequency content consistent with stick shaker activation was present from 22:16:34.1 EST to 22:16:35.4 EST.

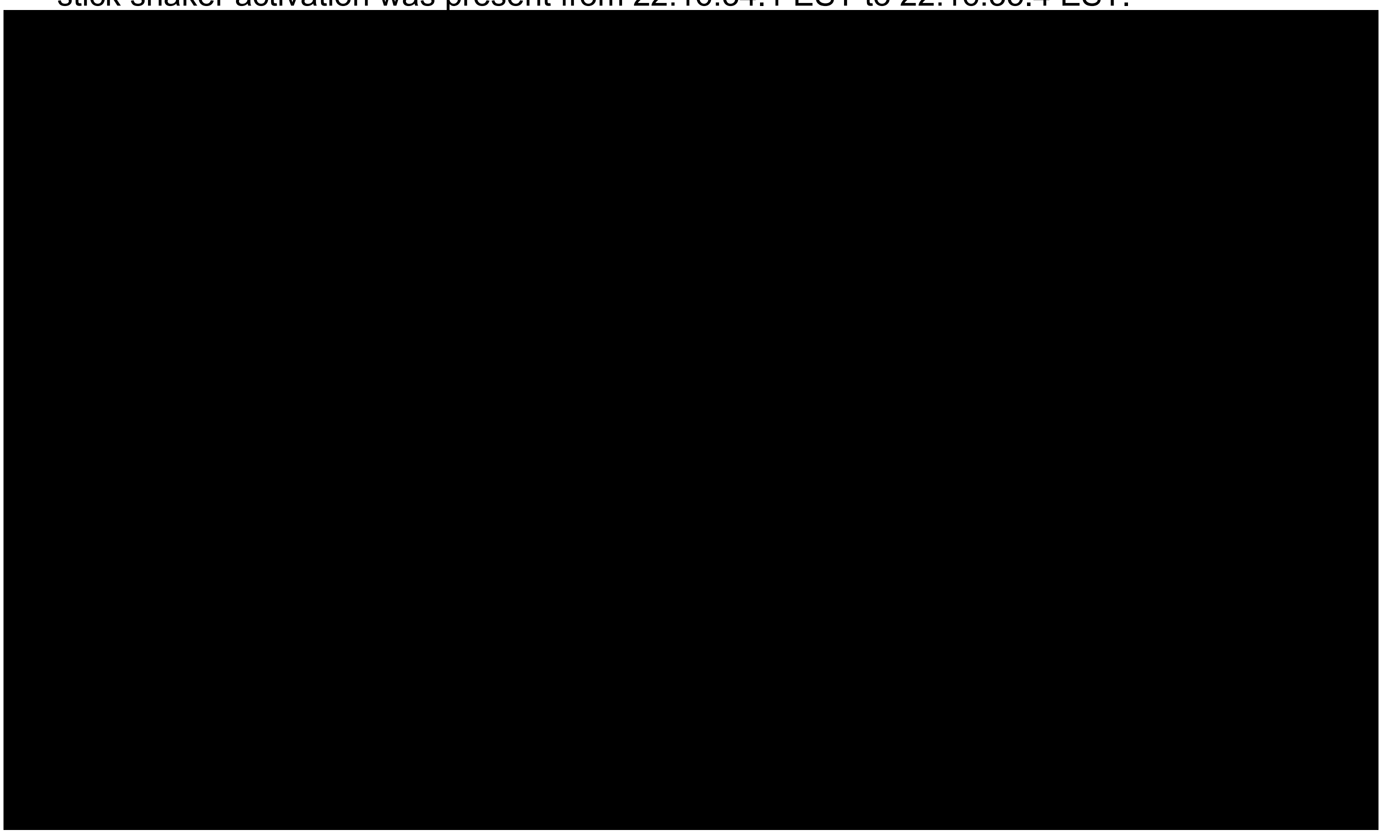

Figure 1. Stick Shaker event during accident flight

Time Analysis

An area of known stick shaker operation was analyzed to determine the frequency content of the stick shaker. Figure 2 shows a spectrogram from the time period 22:16:29.0 EST through 22:16:31.0 EST, an area identified as one of stick shaker activation by the CVR group as well as an area containing an FDR sample indicating the stick shaker was active.

Figure 2. Spectrogram of area known stick shaker activation (time on x-axis in CVR elapsed time [30 min portion])

According to Bombardier:

> *The audible sound associated with stall warning…is produced by a combination of the shaker motor, the mechanical rattle mounted on the control column and control wheel chain slap. Each of these sources is mechanical in nature and subject to slight shifts in frequency, timbre and volume in a prolonged stall warning event.*

The stick shaker operation exhibits a pulse-like broadband frequency response at a very repeatable rattle period of 0.037-0.039 seconds.

[redacted]

Figure 3. Spectrogram of area in question (time on x-axis in CVR elapsed time [30 min portion])

A spectrogram of the time from 22:16:34.1 EST to 22:16:35.4 EST is shown in Figure 3. Note the strong tone present at 1750 Hz; this is a harmonic of the autopilot disconnect tone at a fundamental frequency of 250 Hz and has already been identified by the CVR group. Clearly, the repeatable period of the stick shaker vibration is not immediately obvious during this time; however, upon closer examination there is a small amount of periodic content present, most strongly at approximately 2000 Hz. A qualitative analysis shows that this periodic behavior matches the 0.037-0.039 period of the stick shaker vibration and can be extended back to 22:16:34.4 EST from 22:16:35.4 EST.

Demodulation Analysis

Demodulation is a technique that attempts to examine background noise for any superimposed frequency content modulated by the signal of the stick shaker. The stick shaker period of 0.037-0.039 seconds corresponds to a stick shaker rattle frequency of 25.6-27.0 Hz.

A demodulation filter was applied to three sections of audio: a section where the stick shaker was not active, a section where the stick shaker is known to be active, and the section in question in two parts: 22:16:34.1 EST to 22:16:34.4 EST and 22:16:34.4 EST to 22:16:35.4 EST, the two segments identified in the time analysis. All demodulated spectra are calculated based on a sample size of 8192 with 50% overlap on digital audio captured at 16 kHz.

Figure 4 shows the demodulated sound spectrum of the section of audio where the stick shaker was not active from 22:16:26.1 EST to 22:16:27.1 EST. No frequency content is visible at the 25.6-27.0 Hz frequency of the stick shaker.

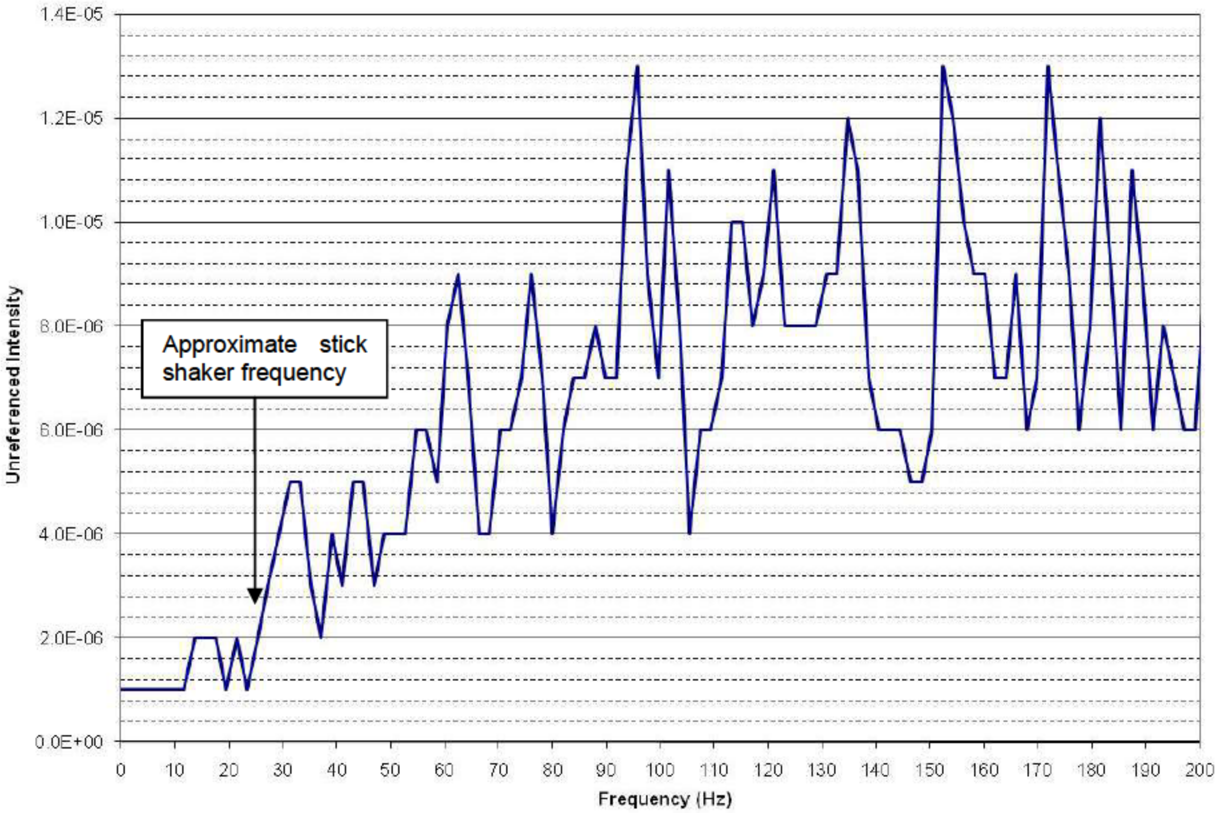

Figure 4. Demodulated spectrum for no stick shaker activation

Figure 5 shows the demodulated sound spectrum of the section of audio where the stick shaker is known to be active from 22:16:31.7 EST to 22:16:32.7 EST. Frequency content at approximately 25.4 Hz is present along with multiple harmonics.

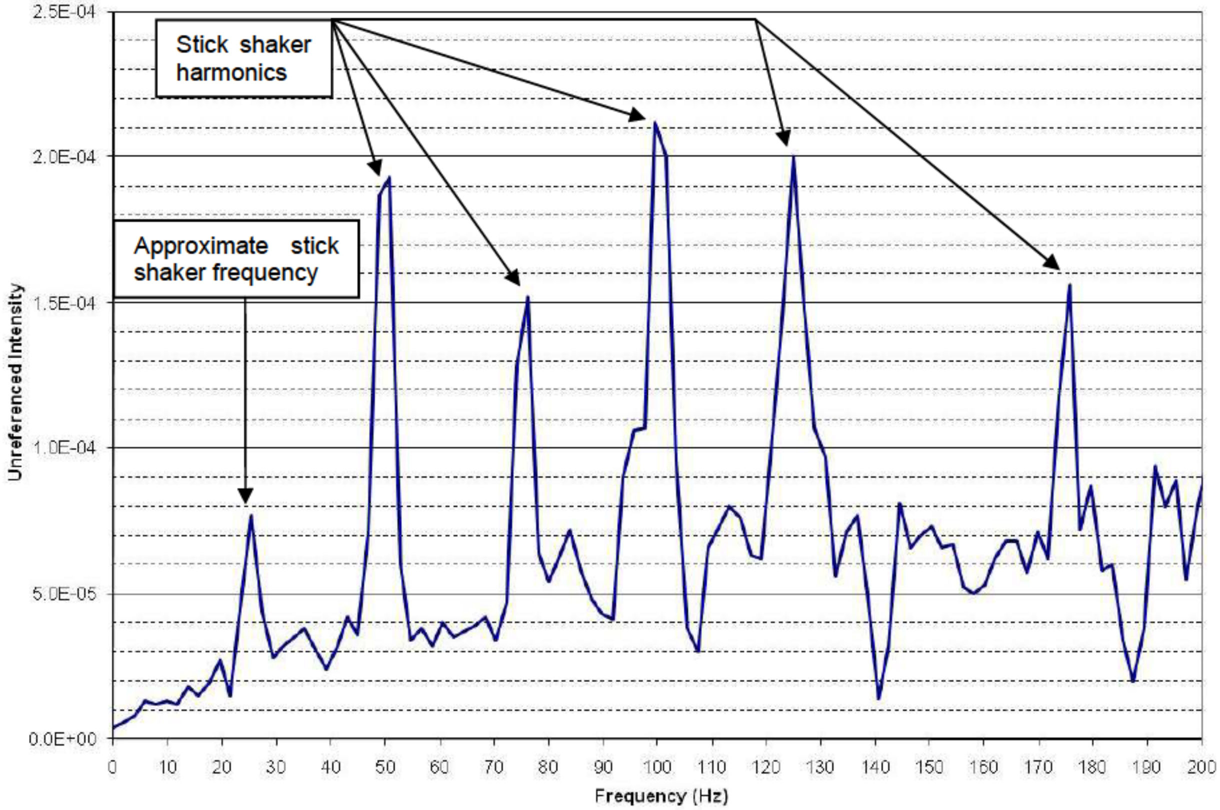

Figure 5. Demodulated spectrum of known stick shaker activation

Figure 6 shows the demodulated sound spectrum of the section of audio between 22:16:34.4 EST and 22:16:35.4 EST identified by the time analysis as having frequency content consistent with stick shaker activation. While not strong, a peak exists above the background noise at 25.4 Hz and multiple harmonics.

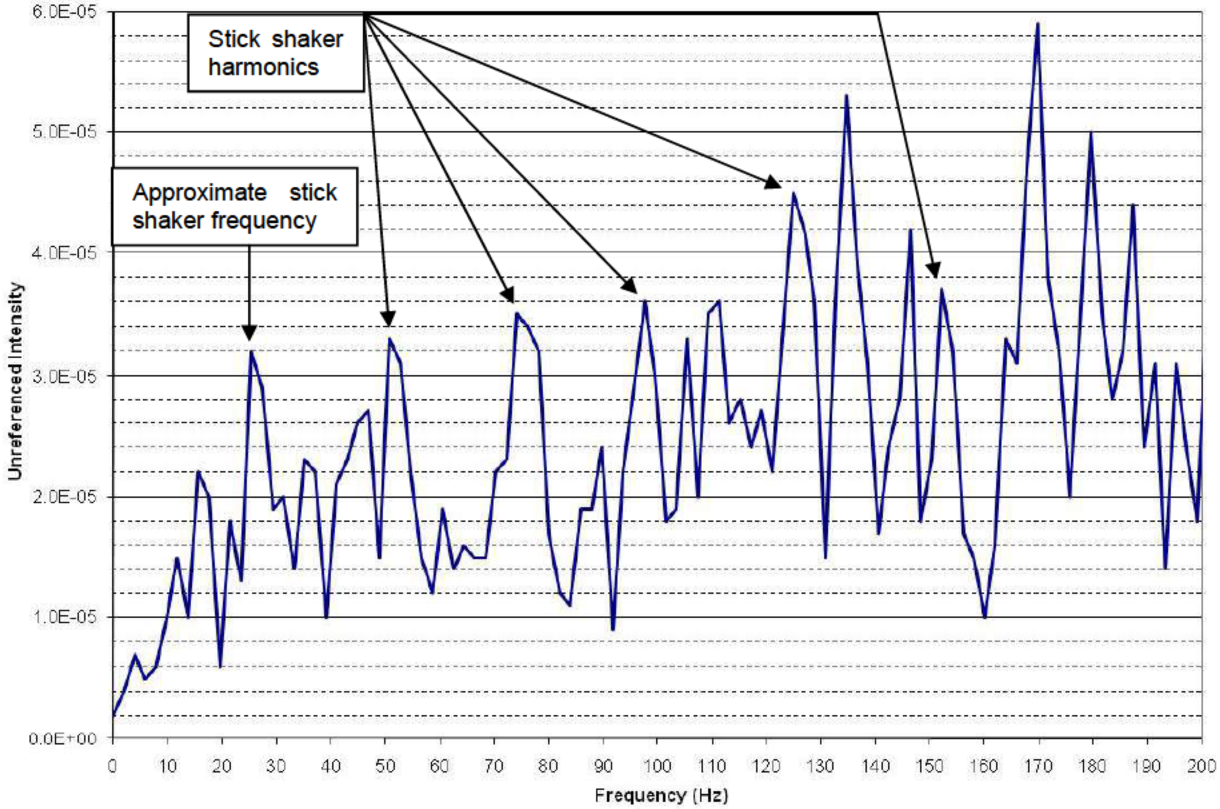

Figure 6. Demodulated spectrum for time 22:15:34.4 EST to 22:16:35.4 EST

Peaks are visible at 21.5 Hz and 31.2 Hz, however no energy present at the measured frequency of the stick shaker was found from 22:16:34.1 EST to 22:16:34.4 EST (Figure 7).

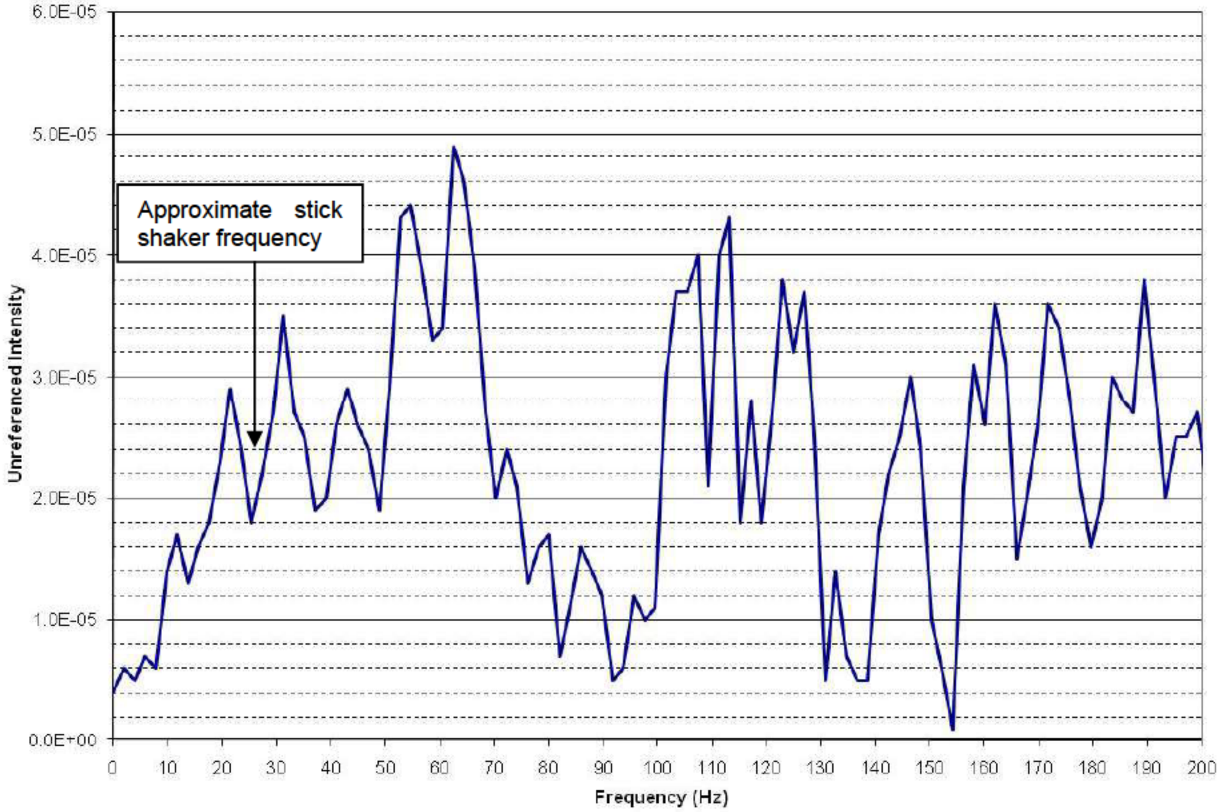

Figure 7. Demodulation spectrum for time 22:16:34.1 EST to 22:16:34.4 EST

Summary

An FDR sample at time 22:16:34.62 EST indicates the stick shaker is active. The CVR group transcript indicates the stick shaker was not active between 22:16:34.1 EST and 22:16:35.4 EST. A sound spectrum study was performed to resolve this inconsistency.

Two separate analyses identified frequency and harmonics consistent with stick shaker activation between 22:16:34.4 EST and 22:16:35.4 EST, however the stick shaker vibration was likely damped due to the lower intensity measured at the stick shaker frequency in both analyses. No frequency content consistent with stick shaker activation was found between 22:16:34.1 EST and 22:16:34.4 EST.

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